Profiling the states out: posterior inference for MAGI without a Gaussian assumption

September 2, 2026

Abstract

MAGI [1] infers ODE parameters by placing a Gaussian process prior on the state and constraining it to satisfy the dynamics on a discretisation grid. Its posterior is dominated numerically by the states, which outnumber the parameters by two orders of magnitude, and the standard summary—a Gaussian at the maximum a posteriori point—is mis-centred by a full posterior standard deviation on a benchmark as benign as FitzHugh–Nagumo.

We argue that the states should not be approximated at all. They are the part of the problem the data and the GP prior constrain most tightly, and integrating them out by Laplace at each parameter value leaves a p-dimensional integral with p between 3 and 7, which can be done directly. The resulting posterior is a mixture, one Gaussian in the states per parameter node, and makes no Gaussian assumption about the parameters whatsoever. On every benchmark in our suite that has a usable reference, its largest parameter error is at or below the level at which two independent halves of that reference agree with each other: 0.0126 against a floor of 0.0100 on FitzHugh–Nagumo, 0.0093 against 0.0081 on HIV, and 0.0119 against 0.0405 on the chaotic Lorenz system, where the mode alone is 1.80 out. It costs between three and twenty-three seconds against roughly two hundred and up for a reference chain, and it reports its own reliability without one.

Getting there required correcting several things upstream, of which the largest was not in the inference at all. MAGI's GP hyperparameters are fitted by an optimiser started at $\phi_1 = \phi_2 = \sigma = 1$ regardless of the data's units; on a benchmark whose states have marginal variance 5.6×10^4 this settles on a lengthscale four orders below the grid spacing, so the GP imposes no smoothness, the trajectory interpolates the observations and is unconstrained between them, and the parameters absorb the difference. Putting that fit on a scale-free footing moves the trajectory error on that system from 69.5% to 0.4% and takes its Hessian condition number from 4×10^{17} to 4×10^2. Separately, the Gauss–Newton normal equations square the Jacobian's condition number, which for parameters in mixed units is repaired by one line of diagonal scaling; the benchmark suite's forward Euler integrator carried more error than the observation noise it was competing with; and MAGI's implementation omits the prior $\pi_\Theta(\theta)$ that its own derivation includes.

We therefore also report a pre-inference diagnosis, costing seconds and no sampling, that establishes whether the mode is a mode, whether it is the only one, which parameters the data determines, whether the posterior is proper, and whether the GP constrains the states at all. The last of these is the one that mattered: the others reported the broken hyperparameter fit faithfully as a property of the model.

1 Introduction

Estimating the parameters $\theta \in \mathbb{R}^p$ of $\dot{x} = f(x, \theta)$ from sparse noisy observations is awkward because the likelihood contains an ODE solve. MAGI [1] removes it: a Gaussian process prior on each state component, a discretisation grid I, and a constraint that the GP satisfy the dynamics on that grid. The unknown becomes

$$x = (\theta, X) \in \mathbb{R}^d, \qquad d = p + nD,$$

and on the four systems we study d runs from 106 to 608 while p runs from 3 to 7.

That imbalance is the whole story of this paper. Almost all of the dimension is state, and the state is the part of the problem that the observations and the GP prior pin down; the parameters are few and are what the exercise is for. Approximating everything at once with a Gaussian at the mode—the default summary—spends its accuracy in the wrong place, and we show it is mis-centred by a full posterior standard deviation on the parameters even when the covariance is nearly right, so nothing in the output looks wrong. Sampling the joint posterior properly is possible but costs minutes and, as we document, can fail silently on exactly the problems where it is most needed.

Contributions.

1. A posterior approximation that profiles the states out and integrates the parameters directly (Section 6), reaching the reference chain's own agreement floor on every benchmark that has a usable reference, in three to twenty-three seconds, and reporting its reliability without a reference.
2. A Gaussian-process hyperparameter fit that is scale-free (Section 4). MAGI's optimiser starts at $\phi_1 = \phi_2 = \sigma = 1$ whatever the data's units, and on states with marginal variance 5.6×10^4 it converges to a lengthscale four orders below the grid spacing, so the prior imposes no smoothness at all. This was the largest error we found, it was upstream of everything else, and it invalidated several of our own earlier conclusions.
3. A pre-inference diagnosis built from the mode and its exact Hessian (Section 5): mode validity, globality, identifiability, properness, whether the GP constrains the states, and how badly conditioned the geometry a sampler would face is—in seconds and without sampling.
4. The observation that MAGI's implementation omits the $\pi_\Theta(\theta)$ term of its own derivation (Section 3), together with a retraction of the demonstration we had built on it, which turned out to be measuring the hyperparameter bug of Section 4.
5. Two corrections without which none of the above is measurable: diagonal scaling of the Gauss–Newton normal equations (Section 2), and a benchmark integrator whose error is below the observation noise (Section 7).
6. Negative results, including that the local curvature of the profiled marginal is provably equal to the joint Laplace's, so profiling buys nothing near the mode (Section 8).

This paper continues [2], which established the exact least-squares structure of the MAGI posterior and a Gauss–Newton mode solver, and proposed a third-order correction to the Laplace mean. That correction is superseded here: it helped on one of the four systems and harmed two, and the profiled posterior of Section 6 dominates it everywhere. Those comparisons were made before the hyperparameter fit, so they describe a different set of posteriors; we have not re-run them, because the correction is not the recommended method under either.

	$\kappa(A)$	$\kappa(DAD)$	unscaled $\|\nabla \log p\|$	scaled $\|\nabla \log p\|$	unscaled s	scaled s
fn	4.4e3	3.9e3	2.2e-07	2.8e-07	2.00	**1.31**
hes1	2.2e3	1.0e2	**3.8e-12**	3.9e-09	1.82	**1.21**
hiv	**4.1e17**	**8.4e10**	**1.7e-02**	**6.3e-09**	3.40	**2.38**
lorenz	9.6e4	7.5e3	4.1e-09	4.3e-08	1.77	**1.26**

Table 1: Symmetric diagonal scaling of the Gauss–Newton solve. Seven orders of magnitude where the parameters are in mixed units; slightly worse where they are not, though every value there is far below any tolerance a user would set.

2 The MAGI posterior and its mode

2.1 Least-squares structure

With the observation noise held fixed, MAGI's negative log-posterior is exactly a sum of squares [2]. Writing $L_c L_c^\top = C^{-1}$, $L_k L_k^\top = K^{-1}$, $b = \beta^{-1}$ for the prior tempering, and P for the parameter prior precision introduced in Section 3,

$$-2 \log p(x) = \|R(x)\|^2 + \text{const}, \qquad R(x) = \begin{bmatrix} \sqrt{b}\, L_c^\top (X - \mu) \\ (X - y)/\sigma \;\text{ on observed entries} \\ \sqrt{b}\, L_k^\top \big(f(X, \theta) - \dot{\mu} - m(X - \mu)\big) \\ P^{1/2}(\theta - \theta_0) \end{bmatrix}. \tag{1}$$

Only the third block is nonlinear and only through f, so $\nabla^2 R_a = \sqrt{b}\,(L_k^\top)_a \nabla^2 f$ pointwise, and the exact Hessian $H = J^\top J + \sum_a R_a \nabla^2 R_a$ costs one Jacobian assembly rather than d autodiff passes. The fourth block is linear, so a parameter prior —including a full precision matrix—leaves this structure untouched.

2.2 The normal equations square the condition number

The solver forms $A = J^\top J$ and factorises it, which is far faster than a QR least-squares solve and squares $\kappa(J)$. Whether that matters is not a property of the science but of the units. Table 1 shows what happens on HIV, whose parameters $\theta = (36, 0.108, 0.5, 1000, 3)$ are rate constants on different scales: the θ-block curvature spans 3×10^{-6} to 5×10^{2}, $\kappa(A)$ reaches 4×10^{17}, and the Cholesky is meaningless.

The repair is symmetric diagonal scaling,

$$D = \operatorname{diag}(A)^{-1/2}, \qquad (DAD + \lambda I)\, s = -Dg, \qquad \delta = Ds, \tag{2}$$

identical in exact arithmetic and completely different in floating point. Three details cost more effort than the change. First, the existing damping was a uniform ridge $\lambda\, \operatorname{tr}(A)/d \cdot I$, not Marquardt's $\lambda \operatorname{diag}(A)$; on a unit-diagonal matrix λI *is* the latter, so scaling makes the damping per-coordinate as a side effect. Second, λ becomes relative, so the previous absolute floor of 10^{-12} sits at the smallest eigenvalue of the scaled matrix and never stops perturbing the solve; machine epsilon is the meaningful floor. Third, and dominant, the *initial* λ matters: at 10^{-8} HIV stalls at 1.5×10^{-3}, while at 10^{-10} or below it reaches 10^{-7}, and the other three systems are indifferent across six orders. A diagonal floor set relative to the largest entry is tempting and wrong—HIV's legitimate curvature range spans 10^{16}, so any relative floor large enough to act clips real curvature.

2.3 Convergence cannot be tested on the gradient norm

The same reasoning applies to stopping the iteration. $\|\nabla \log p\|$ carries the units of the log-density and of θ, so no absolute tolerance transfers between problems, and its attainable floor is set by the working precision: in float32 it bottoms out near 10^{-2} on FitzHugh–Nagumo, where the default tolerance of 10^{-6} is unreachable by construction and the solver runs every iteration of its budget for nothing.

The Newton decrement is the scale-free replacement and costs nothing, since $\delta = A^{-1}g$ is already formed at each step:

$$\lambda_{\mathrm{N}} = \sqrt{g^\top A^{-1} g} = \sqrt{g^\top \delta}. \tag{3}$$

To second order this is the remaining distance to the mode in posterior standard deviations, so a tolerance on it means the same thing on every problem and in every precision. A tolerance of 10^{-3}—locate the mode to a thousandth of a standard deviation—terminates every case in the suite in 6 to 35 iterations where an absolute tolerance ran to the cap:

	fn		hes1		hiv		lorenz	
	f64	f32	f64	f32	f64	f32	f64	f32
iterations	20	22	14	14	6	9	35	31
λ_{N}	1.1e-3	5.6e-3	4.5e-9	4.2e-6	1.2e-5	∞	4.7e-4	1.6e-3
$\|\nabla \log p\|$	2.7e-3	1.5e-2	3.9e-9	7.9e-6	1.9e-3	2.1e0	2.1e-4	8.9e-4

A scale-free tolerance still has a precision floor, so the iteration also stops on stagnation; otherwise float32 spends its remaining budget whenever the requested tolerance happens to sit below what the format can deliver. The entry $\lambda_{\mathrm{N}} = \infty$ for HIV in float32 is not a failure to converge—the mode it returns matches the float64 one to three decimals in $\log p$—but a refusal to certify: at $\kappa(A) = 4 \times 10^{17}$ in single precision the decrement cannot be computed reliably, and reporting that is more useful than reporting a number.

Remark 1 (A recurring theme). This is the fourth place in this paper where a quantity with units was being compared against an absolute threshold, after the conditioning of the normal equations (Section 2), the detection of flat directions (Section 5) and the finite-difference step (Section 6.4). In each case the fix is to divide by the relevant scale—the diagonal of A, the diagonal of H, the posterior standard deviation—and in each case the unscaled version worked on the benchmark it was developed on and failed elsewhere.

3 The missing prior

MAGI's posterior, as derived in [1], is

$$p(\theta, x(I) \mid \cdot) \ \propto\ \pi_\Theta(\theta)\, \exp\Big\{ -\tfrac{1}{2} \textstyle\sum_d \big[\mathrm{GP} + \mathrm{observation} + \mathrm{ODE}\big] \Big\},$$

with “a general prior distribution $\pi(\cdot)$ on θ”. The implementation omits π_Θ: its `theta_conf` argument reaches only the initialiser, where it pulls the starting point toward a guess, and never enters the log-density. Every posterior therefore carries a flat improper prior on every parameter.

By (1) a Gaussian prior is one more residual block, so restoring it preserves the least-squares structure, contributes a constant to $J^\top J$, adds no second-derivative term, and lifts the θ block away from singularity. We take the root from an eigendecomposition rather than a Cholesky because P is permitted to be singular, $P = 0$ recovering the previous behaviour bit for bit.

3.1 What we thought it fixed

We originally reported a striking demonstration here. Under the flat prior HIV's posterior was improper in λ—the profiled log-density fell by 0.07 nats while λ moved from -11.9 to 99,988—the Hessian carried an exact null direction, importance sampling collapsed to an effective sample size of one, and the parameters that *were* well determined, δ and N, sat at 0.259 and 1788 against a truth of 0.5 and 1000 with posterior standard deviations under 2% of their values. Supplying a proper prior removed the null direction, made the posterior proper, restored the effective sample size fourteenfold and moved δ and N to 0.481 and 962. A control with the prior centred at half the true values still recovered 0.470 and 940, so the prior was not supplying the answer.

All of that was measuring the hyperparameter fit of Section 4. With ϕ fitted on a scale-free footing, HIV's posterior is proper under a flat prior, has no null direction, and places λ at 36.3 against a truth of 36. The demonstration is void.

What survives is narrower but still worth stating. The implementation omits a term its own derivation contains, and that is a discrepancy independent of any benchmark: wherever a parameter genuinely is unidentified, a flat prior leaves the posterior improper, and the machinery for supplying π_Θ costs one residual block and preserves the least-squares structure exactly. The mechanism we described—that in a partially identified model the estimates of the identified parameters are set by wherever the unidentified ones come to rest—remains sound as an argument, but we no longer have an example of it on this suite, because the suite turns out to be identified.

Remark 2. The episode is worth keeping in view when reading the rest of this paper. Every diagnostic we had built reported the situation correctly given the model it was handed; what none of them could do was question the model. The observation that broke it open was not a diagnostic at all but a plot of the fitted trajectories.

4 The hyperparameters the inference never sees

MAGI fits the GP hyperparameters $\phi = (\phi_1, \phi_2)$ once, before any inference, by maximising a marginal likelihood per state component. Nothing downstream revisits them, and nothing in the posterior machinery can compensate for them. They turn out to dominate everything else in this paper.

4.1 The symptom

The posterior trajectories are wrong on three of the four systems, and not in a way the parameter diagnostics detect. On HIV's T_U, whose observation noise is $\sigma = 3.16$ and whose true range is 168 to 600:

	rms/σ
MAP against the data, at observed points	0.007
MAP against the truth, at observed points	1.06
MAP against the truth, at **unobserved** points	**74.5**

The trajectory passes exactly through every observation and falls to 9.5×10^{-8} between them. Observations occupy every second grid point, so the interleaved points are held by the GP prior alone—and the fitted lengthscale is 1.05×10^{-4} against a grid spacing of 0.1. The fitted posterior is not at fault: its trajectory bands match the reference chain to within a few percent on every system. The mode is.

	before		after	
system	$\ell/\delta t$	trajectory error	$\ell/\delta t$	trajectory error
fn	13, 12	3.5%, 7.2%	13, 12	3.5%, **2.6%**
lorenz	12, 9.1, 1.1	22.9%, 25.3%, 14.2%	12, 9.1, 17	**8.3%, 11.8%, 5.3%**
hes1	0.14, 0.15, 0.16	71.6%, 73.2%, 100.8%	8.0, 6.3, 3.9	**5.8%, 13.8%, 30.0%**
hiv	0.001, 0.011, 12	69.5%, 6.4%, 0.3%	50, 50, 8.7	**0.4%, 0.9%, 0.1%**

Table 2: Relative error of the fitted state trajectory against the truth, before and after the hyperparameter fit is made scale free. Wherever the lengthscale exceeds a few grid spacings the trajectory is recovered; wherever it falls below one, the states between observations are held by nothing.

4.2 The cause, and a scale-free fit

The hyperparameters are optimised in $\log\phi$ by BFGS from $x_0 = 0$, that is from $\phi_1 = \phi_2 = \sigma = 1$ whatever the units of the data. HIV's T_U has marginal variance 5.6×10^4, so the start is five orders out, and BFGS—whose step sizes and convergence tolerances are absolute in its own variables—settles on a lengthscale that explains the observations as white noise.

The repair is the same one as everywhere else in this paper: optimise $\phi = s \odot e^u$ from $u = 0$, with s the data's own marginal variance and the FFT lengthscale estimate the code already computes. A unit step is then a factor of e regardless of units.

That alone is not enough, because the constraint is two-sided. A scale-free start pushes HIV's T_I lengthscale to 20.56, the entire time span, at which K^{-1} becomes *indefinite* (eigenvalues -24 to $+11$) and its Cholesky in the solver is NaN: the Matern derivative kernel degenerates as the lengthscale approaches the span, just as it stops constraining anything when the lengthscale falls below the grid. The fit is confined to $[2\,\delta t,\ \mathrm{span}/4]$, both ends being properties of the discretisation rather than of the data.

Trajectory quality is governed entirely by the resulting ratio $\ell/\delta t$ (Table 2), and `diagnose`() now reports it, because no other diagnostic can see it.

4.3 What this invalidates

The consequences reach most of the quantitative claims we had made about these benchmarks. With the hyperparameters fitted properly, **every system recovers its true parameters to within** 1.6 **posterior standard deviations**, and the only parameters still weakly determined anywhere are three of Hes1's seven—HIV's λ is 36.3 against a truth of 36 where it had been -11.9, and Hes1's seven parameters, which had collapsed to $\sim 10^{-6}$, are all within 1.4 standard deviations of their true values. HIV's condition number falls from 4×10^{17} to 4×10^{2} and its null direction disappears, so its posterior is proper and a reference chain for it exists.

We record this rather than quietly rewriting, because the failure mode is the instructive part. The bug sat upstream of everything under study; it produced posteriors that looked entirely plausible; and the diagnostics of Section 5—mode validity, identifiability, properness— reported it faithfully, as a property of the model. What exposed it was plotting the trajectories, which none of those diagnostics examined.

5 Diagnosis before inference

A fast mode makes a set of questions affordable that are usually assumed away. All of the following requires no sampler and costs, on these four systems, between ten seconds and a minute and a half — most of it in the eight Hessian factorizations of the conditioning check below, which dominates on HIV at $d = 608$ and can be dropped by setting `n_curv=0` when only identifiability is wanted.

Is it a mode? The exact Hessian's spectrum, and, if a genuinely negative direction exists, a step along it followed by a re-solve. On all four systems there is none. Proximity to the mode is reported as $\sqrt{g^\top H^{-1} g}$ rather than $\|\nabla \log p\|$, for the reason in Section 2.3: on FitzHugh–Nagumo in float32 the raw norm is 1.7×10^{-2}, which fails any tolerance carried over from double precision, while the scale-free measure is 5.6×10^{-3}—the mode is located to within 0.6% of a posterior standard deviation, and nothing downstream is affected.

Is it the only one? Re-solving from dispersed starts, and separately by continuation in the prior tempering β^{-1}: at small β^{-1} the objective is dominated by the observation term and has a benign landscape, and following the solution up to the target keeps the iterate in that basin. Two solves count as having found the same optimum when the POINTS agree, in the Hessian metric, and not when their log-densities agree to some number of decimals—in float32 the mode is located only to a few thousandths of a standard deviation, so $\log p$ between two solves of the same optimum differs by far more than any tight absolute tolerance, and comparing it reports every restart as a new optimum. On this criterion FitzHugh–Nagumo, Hes1 and HIV each return a single optimum over five solves. Lorenz returns three, none of them better than the mode, which is the expected signature of a chaotic objective with shallow secondary basins; the check passes on the grounds that the mode found is still the best one, and the count is reported so that the distinction is visible rather than hidden behind the verdict.

Which parameters are identified? Eigenvalues of the raw Hessian have units, and reading flatness off them is the same scale error that Section 2 is about. We use the spectrum of DHD and the marginal posterior standard deviation of each parameter as a fraction of its own value.

Will it be hard to sample? The Laplace metric whitens the target exactly *at* the mode, so what matters is how far it departs a standard deviation away. Writing $M = L^\top H(x) L$ for L the mode's whitening, $\mathrm{cond}(M)$ measured over a handful of Laplace draws predicts the reference chain's behaviour before any chain is run (Table 3), and costs about eight Hessian evaluations. It is not visible in the condition number at the mode: on Hes1 the latter is 1.5×10^5 while $\mathrm{cond}(M)$ reaches 10^6.

Is the posterior proper? A null direction of the quadratic model is not an improper posterior; the density may still decay at higher order. We walk each parameter axis outward with the states re-profiled at every step, so the walk follows the ridge rather than cutting across it, and record the fall in $\log \hat{p}$. The walk is repeated at several radii and the fall reported at the furthest one that resolved, since a probe whose inner solve fails also returns $-\infty$ and would otherwise be indistinguishable from a density that has genuinely decayed. With the hyperparameters fitted correctly every parameter of every system is proper, and on three of the four the probe stops resolving before the density measurably decays—so the test now confirms properness without measuring how strongly, which is weaker than it looks and worth remembering if it is relied on.

system	cond(M), Laplace draws	reference draws	reference $\widehat{R}$	divergences
hiv	**1.21**	1.30	**1.0001**	**0%**
lorenz	1.9×10^2	2.2×10^3	1.0072	1.1%
fn	3.8×10^2	1.7×10^3	1.0064	1.4%
hes1	$\mathbf{1.1 \times 10^6}$	7.1×10^6	**1.76**	**13%**

Table 3: Variation of the curvature over the posterior, and the reference chain that resulted. Laplace draws reproduce the ordering obtained from reference draws, so the difficulty is knowable in advance.

system	parameter	sd/$\lvert\hat{\theta}\rvert$	verdict
fn	a, b, c	0.13, 0.38, 0.02	identified
hiv	$\lambda, \rho, \delta, N, c$	0.04, 0.06, 0.02, 0.02, 0.01	identified
lorenz	β, ρ, σ	0.06, 0.03, 0.12	identified
hes1	b, c, d, e	0.11 − 0.26	identified
hes1	a, f, g	0.50, 0.56, 0.53	weak

Table 4: Identifiability after the hyperparameter fit of Section 4, in five to twelve seconds per system without sampling. Every system recovers its true parameters to within 1.6 posterior standard deviations. Before the fix this table read very differently—HIV's λ improper, all seven of Hes1's parameters diffuse—and none of that was a property of the models.

Each check is reported with the threshold it must meet and a pass or fail, alongside an overall status, so that "the method declined" and "the posterior does not exist" are not left for the reader to disentangle. Hes1 is the case where the two reports disagree, and they disagree in the direction that matters: the *diagnosis* fails, on curvature variation alone—a unique maximum, no negative curvature, every parameter proper, but cond(M) $= 1.1 \times 10^6$—while the *fit* passes its gate, at ESS $= 21\%$ and $\widehat{k} = -0.06$. Both are right. The profiled integral never sees the funnel, because the states are profiled out and only seven dimensions are sampled; the reference chain has to traverse it, and cannot. The practical consequence is uncomfortable and worth stating plainly: on Hes1 we have an estimate that certifies itself and no way to check it, because the thing we would check it against is the thing the diagnosis says is infeasible.

One gap in the gate is visible here. On Hes1 93 of the 512 inner profile solves fail outright, and the gate does not look at that count—it reads the importance weights, which are computed from the 419 nodes that succeeded. A high failure rate is not the same as a bad weight distribution and should probably be gated separately; we report it in the diagnostics but do not currently act on it.

6 The profiled posterior

6.1 Construction

Do not approximate the states. Integrate them:

$$p(\theta) = \int e^{-U(\theta,X)}\,dX \;\approx\; e^{-U(\theta,X^\star(\theta))}\,\det H_{XX}\big(\theta, X^\star(\theta)\big)^{-1/2}, \qquad X^\star(\theta) = \arg\min_X U(\theta, X), \tag{4}$$

the Laplace approximation applied to the *inner* integral only. The remaining p-dimensional integral is done by self-normalised importance sampling on a scrambled Sobol set, so the result is deterministic given a seed and comes with an effective sample size and a Pareto $\widehat{k}$ as reference-free diagnostics.

This differs from a Gaussian at the joint mode in three ways that matter. It is exact under Condition 1 below, where the joint version still carries a first-order mean error, because the joint version linearises the θ–X coupling and this one does not. What remains is the non-Gaussianity of $p(X \mid \theta)$ alone, which the GP prior and the data constrain far more tightly than they constrain θ. And nothing Gaussian is assumed about θ: skew and curvature in the parameter marginals survive.

The output is not a Gaussian but a mixture, one Gaussian in the states per parameter node,

$$p(x) \;\approx\; \sum_i w_i\, \delta(\theta - \theta_i)\, \mathcal{N}\big(X;\, X^\star(\theta_i),\, H_{XX}(\theta_i)^{-1}\big),$$

from which every moment follows in closed form, including $\mathrm{Cov}(\theta, X)$, which no Gaussian centred at the mode gets right.

Condition 1 (Affine in the state)**.** $f(\cdot, \theta)$ is affine in X for every θ.

Proposition 1. *Under Condition 1 the residual* (1) *is affine in X at fixed θ, so $U(\theta, X)$ is exactly quadratic in X, $p(X \mid \theta)$ is exactly Gaussian, and* (4) *holds with equality. The d-dimensional problem is then exactly a p-dimensional one.*

Condition 1 is much weaker than requiring f affine in (X, θ) jointly—θ may still multiply states—and covers linear compartment models, linear pharmacokinetics and every constant-coefficient system. It is testable in two Hessian evaluations by checking whether H_{XX} depends on X [2].

6.2 The proposal, and a structural limit

Drawing θ from the joint Laplace marginal $\mathcal{N}(\hat{\theta}, (H^{-1})_{\theta\theta})$ is the obvious choice and a poor one: that object accounts for the θ–X coupling only to first order, and on Hes1 the log importance weights then span 8,400 nats with ESS = 0.6%. Since p is small, the profiled marginal's own mode and curvature are directly computable—a Newton solve in p dimensions by central differences through (4), costing $O(p^2)$ profile solves per step.

This is worth doing for the mode. The profiled mode is *not* the joint mode's θ: on FitzHugh–Nagumo it sits $(-0.24, +1.01, -1.09)$ Laplace standard deviations away, precisely the directions and magnitudes by which the joint mode is known to be biased. The joint mode maximises $U(\theta, X)$; the profiled mode maximises $U(\theta, X^\star(\theta)) + \frac{1}{2}\log\det H_{XX}(\theta)$, and that determinant—the volume of state space consistent with θ—is what the joint mode ignores.

It is *not* worth doing for the curvature, for a structural reason. The second derivative of $-U(\theta, X^\star(\theta))$ at the mode is exactly the Schur complement $(\Sigma_{\theta\theta})^{-1}$, and the determinant term's curvature is negligible: measured on Hes1 at three stencil sizes spanning two orders of magnitude, the profile curvature equals the joint Laplace curvature to four decimals. Profiling buys nothing near the mode; everything it gains is in the shape away from it.

6.3 A warm start from the implicit function theorem

$X^\star(\theta)$ satisfies $\nabla_X U(\theta, X^\star) = 0$, so $H_{XX}\, dX^\star/d\theta + H_{X\theta} = 0$ and the sensitivity is one solve against a factorisation already in hand. Starting each node from $X_{\mathrm{MAP}} + (dX^\star/d\theta)(\theta - \theta_{\mathrm{MAP}})$

inner iterations	2	4	8
from X_{MAP}	0.0437 / 5.8 s	0.0065 / 6.8 s	0.0056 / 8.7 s
from the predictor	**0.0059 / 4.5 s**	0.0063 / 6.1 s	0.0065 / 9.1 s

Table 5: Largest parameter error in reference standard deviations, and wall clock, on FitzHugh–Nagumo. The chain-to-chain floor is 0.0120.

	h/sd	0.05	0.2	0.8	1.6	3.2	6.4	12.8
fn, float64	θ_1	1.145	1.145	1.150	1.163	1.217	1.425	—
fn, float32	θ_1	**0.439**	1.147	1.146	1.162	1.217	1.425	—
hes1, both	θ_5	169.4	129.0	**34.5**	13.0	4.0	1.3	—
lorenz, both	θ_2	0.505	0.504	0.482	—	—	—	—

Table 6: Diagonal curvature of the profiled log-density, scaled so a Gaussian profile would read 1.0 at every h. Three different failure modes bound the step: round-off at small h in single precision (fn), truncation at large h where the profile is non-quadratic (hes1), and outright failure of the inner solve (lorenz). The usable windows do not overlap—fn in float32 needs $h \gtrsim 0.2$ while hes1 needs $h \lesssim 0.05$—so no constant serves both.

rather than from X_{MAP} lets the inner iteration count fall from six to two at unchanged accuracy (Table 5).

6.4 Choosing the finite-difference step

The Newton step of the previous subsection differentiates (4) by central differences, and the step size h is not a free parameter to be set once. A central second difference carries truncation error growing like h^2 times the fourth derivative and round-off error falling like σ/h^2, so h is bounded on both sides, and *both bounds are properties of the problem*: σ comes from the working precision, the fourth derivative from how non-Gaussian the profiled marginal is. Table 6 shows the two bounds measured across the suite, in units of the joint Laplace standard deviation.

Since a plateau is precisely a range over which the estimate does not move, we locate it: evaluate the diagonal curvature along a ladder, take the longest *contiguous run of rungs that all agree with one another* to within a tolerance, and use the geometric middle of that run. Three details are each necessary, and we record them because the obvious version of the rule fails on this suite in three different ways.

Agreement must be across the run, not between neighbours. On FitzHugh–Nagumo every consecutive change stays under 5% from $h = 0.05$ out to 3.2 while the curvature drifts from 1.145 to 1.217; small steps accumulate, and a neighbour test walks steadily off the plateau it is meant to find.

The middle of the run, not the top. The ends of the run are exactly where the estimate begins to fail, and both bounds are multiplicative, so the geometric middle maximises the margin on either side. Taking the top is measurably worse: the run is located from diagonal probes while the Newton also needs off-diagonal ones at $\sqrt{2}$ times the displacement, and on Lorenz the top of the run made those non-finite and cost three quarters of the effective sample size.

A failed probe is not agreement. Judging coherence on whichever parameters happen to be finite silently drops those that failed. On Lorenz the third parameter stops resolving beyond $h = 0.8$, and ignoring that extended the apparent plateau to 3.2—the same collapse again, by a

Algorithm 1: Profiled MAGI posterior, $\approx 5\,\mathrm{s}$ at $d = 325$

Input: MAGI model; nodes $N = 512$; inner iterations 3; gate (ESS $\geq 10\%$, $\widehat{k} < 0.7$)
Output: profiled mixture, or Laplace fallback, with diagnostics

$x_\star \leftarrow$ Jacobi-scaled Gauss–Newton on $\|R\|^2$ `// ≈ 1 s`
$H \leftarrow J^\top J + \sum_a R_a \nabla^2 R_a$; Jacobi-stabilised $\Sigma \leftarrow H^{-1}$
$dX^\star/d\theta \leftarrow -H_{XX}^{-1} H_{X\theta}$ `// implicit function theorem`
$h \leftarrow$ geometric middle of the curvature plateau `// Sec. 6.4`
$\hat{\theta}, H_{\text{prof}} \leftarrow$ Newton on $\log \hat{p}(\theta)$, central differences at h
$\theta_i \leftarrow \hat{\theta} + L z_i$, z_i scrambled Sobol, $LL^\top = H_{\text{prof}}^{-1}$
foreach θ_i (batched) **do**
  $X \leftarrow X_{\text{MAP}} + (dX^\star/d\theta)(\theta_i - \theta_{\text{MAP}})$
  three Gauss–Newton steps on X at fixed θ_i
  $\log \hat{p}_i \leftarrow -U(\theta_i, X) - \frac{1}{2} \log\det H_{XX}$
end
$w \propto \exp(\log \hat{p}_i - \log q_i)$; compute ESS and $\widehat{k}$
return *mixture if the gate passes, else* $\mathcal{N}(x_\star, \Sigma)$

different route. A rung counts only if every parameter resolved on it.

The cost is $2p$ evaluations per rung, negligible beside the node budget. The selected steps differ by more than an order of magnitude across the suite (Table 8), which is the case for selecting rather than setting them.

6.5 A gate

Importance sampling at a few percent effective sample size is not an estimate. We require ESS $\geq 10\%$ and $\widehat{k} < 0.7$, both computable without a reference, and otherwise report the Laplace approximation. This is not a formality: on Hes1, where no parameter is identified, the mode's parameter error is 0.0126 against a floor of 0.0181—the mode is already at the reference mean—while the profiled estimate is substantially worse. Where nothing is identified the Laplace approximation is the correct answer and the method must say so.

7 Benchmarks

7.1 Systems, and a note on constructing them

Four systems: **FitzHugh–Nagumo** ($d = 325$, $p = 3$), **Hes1** ($d = 106$, $p = 7$, a rational nonlinearity, $\exp(X)$ states and a component never observed), **HIV** ($d = 608$, $p = 5$, time-dependent forcing, states spanning 30 to 10^5) and **Lorenz** ($d = 306$, $p = 3$, chaotic).

The suite originally integrated with forward Euler, retaining every substep. Both choices bound what could be measured. Storage is $(t_{\max}/h)\,D$, which at $h = 10^{-6}$ is 5.8 GB for Hes1's horizon of 240; and on HIV the integration error at the observation times was 2.6 observation standard deviations, larger than the noise it was competing with, so the posterior under study was not the intended one. Replacing Euler with RK4 and retaining only the output grid changes both:

system	estimate	max $|\theta$ error$|$	max $|$sd error$|$	s
fn	mode	1.0286	2.18%	—
	profiled	**0.0126**	**0.45%**	6.7
	reference half-vs-half	0.0100	0.86%	
hiv	mode	0.1485	0.62%	—
	profiled	**0.0093**	**0.34%**	23.2
	reference half-vs-half	0.0081	0.77%	
lorenz	mode	1.8026	39.68%	—
	profiled	**0.0119**	**1.27%**	5.2
	reference half-vs-half	0.0405	1.19%	
hes1†	mode	2.8639	189.6%	—
	profiled	0.2163	50.5%	3.2
	reference half-vs-half	0.0841	10.9%	

Table 7: Parameter errors in reference standard deviations, after the hyperparameter fit of Section 4. On FitzHugh–Nagumo, HIV and Lorenz the profiled posterior is at or *below* the level at which two independent halves of the reference agree with each other. Lorenz in particular moves from 0.34 before the fix—which we had attributed to chaotic dynamics defeating the inner Laplace approximation—to 0.0119, so that explanation was wrong. †Hes1's reference has $\widehat{R} = 1.76$ and 13% divergences and is not usable; its row is shown for completeness only.

	error / σ		f evaluations	
system	Euler 10^{-6}	**RK4** 10^{-3}	Euler 10^{-6}	**RK4** 10^{-3}
fn	6.8e-05	**6.8e-11**	20,000,000	**80,000**
hes1	2.2e-06	**7.5e-12**	240,000,000	**960,000**
hiv	5.0e-03	**5.2e-09**	20,000,200	**80,800**
lorenz	1.7e-04	**1.1e-08**	2,500,100	**10,400**

Six orders of accuracy for two to three fewer of work, with storage independent of the step: 1 to 5 kB rather than gigabytes. Hes1's integration falls from 10.0 s to 0.14 s. We record this because a benchmark whose synthetic "truth" carries more error than its own noise cannot discriminate between posterior approximations, and the defect is invisible unless looked for.

7.2 References

References are 32 chains of 3,000 draws of NUTS [4] via BlackJAX, preconditioned by the exact Hessian in Jacobi-scaled coordinates, which affects only efficiency and never the stationary distribution. Rebuilt after the hyperparameter fit, HIV gives $\widehat{R} = 1.0001$ with *zero* divergences—it had previously failed to converge at all— FitzHugh–Nagumo 1.0064 with 1.4% and Lorenz 1.0072 with 1.1%. Hes1 gives $\widehat{R} = 1.76$ with 13% divergences and 2.79 million leapfrog steps per chain, and is not usable; it was marginally usable *before* the fix, for reasons Section 10 sets out.

7.3 Results

Table 7 is the main result. On all three systems with a usable reference the profiled posterior sits at or below the level at which two independent halves of that reference agree with each other, which is the most that can be claimed from a finite chain: 0.0126 against a floor of 0.0100 on

system	chosen h/sd		plateau		ESS	
	f64	f32	f64	f32	f64	f32
fn	0.1	0.8	0.01–1.6	0.4–1.6	75.7%	76.4%
hes1	0.02	0.02	0.01–0.05	0.01–0.05	0.4%	0.4%
hiv	0.1	0.4	0.01–1.6	0.1–1.6	2.6%	3.5%
lorenz	0.1	0.4	0.01–0.8	0.2–0.8	75.5%	76.0%

Table 8: Steps selected automatically, and the plateaus they were selected from. The lower edge moves with precision (fn: 0.01 in double, 0.4 in single) because round-off sets it; the upper edge does not (hes1 at 0.05, lorenz at 0.8) because truncation and probe failure set it. The rule reads both without being told which regime it is in.

FitzHugh–Nagumo, 0.0093 against 0.0081 on HIV, and 0.0119 against 0.0405 on Lorenz, the last comfortably *below* it. Against the mode the improvement is 82, 16 and 151 times on location, and the spread errors fall to a fraction of a percent on the first two. The costs in the last column—between three and twenty-three seconds—are against reference chains of roughly two hundred seconds and up.

Lorenz deserves a note, because we previously drew a conclusion from it that was wrong. Before the hyperparameter fit its profiled error was 0.34 against a floor of 4.14, and we attributed the gap to chaotic dynamics making $p(X \mid \theta)$ strongly non-Gaussian and so undermining the inner Laplace approximation (4). That was a plausible mechanism and it was not the one operating: with ϕ fitted properly the error is 0.0119, below the floor, and the chaos costs nothing measurable. Whether (4) degrades on genuinely chaotic problems remains open; this suite no longer speaks to it.

Hes1's row is shown for completeness and should not be read as a result. Its gate passes (ESS = 21%), but its reference does not converge, so 0.2163 is a comparison against a chain that has not found the posterior rather than a measurement of error. Section 10 takes up why.

7.4 Precision, and what the stencil was hiding

The profiled estimate appeared at first to require double precision: in single precision its effective sample size on FitzHugh–Nagumo fell from 75.7% to 2.4% and the gate replaced it with the very approximation it was meant to improve on. That diagnosis was wrong, and the way it was wrong is worth recording.

The importance weights are barely affected by working precision. Measuring the float32 error in $\log \hat{p}$ across profile nodes—and only its *spread* matters, since a constant offset cancels in self-normalisation—gives 0.007 nats on CPU and 0.052 on GPU, which degrade the effective sample size by a factor of 0.997. Guarding the one unguarded matrix–matrix product in the exact-Hessian assembly, which runs in TF32 by default on tensor-core hardware, improves the log-determinant's error spread by a factor of ten on GPU and changes the effective sample size not at all.

What was actually failing was the finite-difference stencil. A central second difference divides by h^2, so at $h = 0.02\,\mathrm{sd}$ the 0.007 nats of round-off become an amplification of roughly 2500 and the curvature estimate is noise—on FitzHugh–Nagumo it comes out *negative*. With the step chosen as in Section 6.4 rather than fixed, single precision matches double everywhere and is consistently faster (Table 8).

We note in passing a fact visible only in these diagnostics: Lorenz's profiled curvature for σ is half the Laplace value, and for ρ four fifths, across the whole plateau. The Laplace covariance is materially wrong there in a way the curvature at the mode cannot reveal, which is the effect profiling exists to capture.

8 Negative results

The third-order correction of [2] is superseded. It helped on FitzHugh–Nagumo and harmed Lorenz (4.14 $\to$ 4.86) and Hes1. Those figures predate the hyperparameter fit and so describe different posteriors; we did not re-measure them, because the profiled estimate is better than either under both.

Correcting the inner Laplace approximation by sampling makes matters worse. Writing the exact inner integral as $-U^\star - \frac{1}{2}\log\det H_{XX} + \log \mathbb{E}_\xi[e^{-\Delta}]$ with $\xi \sim \mathcal{N}(0, H_{XX}^{-1})$, the omitted term can be estimated with antithetic pairs, which cancel Δ's cubic part exactly. It fails: Δ is a sum of quartic terms over 322 to 603 coordinates, a handful of pairs cannot resolve its mean, and injecting that noise into $\log \hat{p}(\theta)$ corrupts the weights multiplicatively. FitzHugh–Nagumo goes from 0.0086 to 0.3665. That it reaches the reference floor with the correction *off* is the useful part.

Restricting the integration to an identifiable subspace did not rescue HIV, when HIV still appeared to need rescuing. Profiling the apparently improper directions out rather than integrating them is well defined—the determinant becomes a pseudo-determinant over the complement—but the effective sample size stayed at 1.6% and half the inner solves failed. We record it because the construction is sound and may be useful on a genuinely partially identified problem; the target it was built for turned out to be an artifact.

Profiling buys nothing near the mode, for the structural reason in Section 6.

Two convergence tests that report success on failure. Clamping the Newton decrement with $\max(g^\top\delta, 0)$, to guard against a negative value, converts a round-off failure into a reported convergence: on HIV in float32 the dot product comes out negative, the clamp returns zero, and the solver stops after one iteration at a point 246 nats below the mode. A non-positive or non-finite decrement must be treated as *not converged*. Separately, counting distinct optima by rounding $\log p$ reports every restart as a new optimum in single precision. Both produced confident, wrong answers rather than visible failures.

A fixed finite-difference step is not safe, and the three natural ways of choosing one adaptively each fail on this suite: a neighbour-agreement test walks off the plateau, the top of the plateau breaks the off-diagonal probes, and treating a failed probe as agreement extends the plateau past its real end. Section 6.4 gives the measurements. We record these because the first two produced plausible-looking answers on three of the four systems and failed only on the fourth.

9 Limitations

The evidence is four systems from one benchmark suite, with usable references on three of them. The profiled posterior reaches the reference floor on all three, which is as much as a finite reference can establish and no more: it bounds the error by the chain's own Monte Carlo noise rather than measuring it. A sharper test would need either a longer reference or a problem with a known answer.

We also no longer have a case in this suite where the inner Laplace approximation (4) visibly degrades. We thought Lorenz was one and it was not (Section 7.3); Condition 1 remains the right diagnostic for it, and remains unrun against a case that fails.

The gate's thresholds (ESS $\geq 10\%$, $\widehat{k} < 0.7$) are conventional rather than calibrated here, and were correct on all four systems, which is weak evidence.

Hes1's reference is unusable, so on one system of four our comparisons have no target; Section 10 sets out why and what might be done about it.

Finally, we argued that the choice of prior on unidentified parameters determines the estimates of the identified ones. The argument stands, but our evidence for it does not: the example was HIV, which is not in fact unidentified. Whether the effect matters in practice is therefore open, and would need a problem that is genuinely partially identified once its hyperparameters are fitted properly.

10 A funnel, and how one might sample it

Hes1 is the one system in the suite without a usable reference, and the reason is worth setting out because it is diagnosable and because the remedies are not the obvious ones.

10.1 What it is

Its local curvature varies by a factor of 35.7 across its own posterior, against 1.01 for HIV. Regressing log of the largest Hessian eigenvalue on the coordinates over reference draws identifies the drivers: the peak of the third state component, correlation $+0.933$, and the parameter a, correlation -0.902 with a slope of -97.6 per unit. Hes1's states are on a log scale, its field reading $P, M, H = \exp(X)$, and a enters as $-aH$ and $-aP$. So a parameter multiplies an exponentiated state, and the state in question, H, is *never observed*. That is a funnel in the textbook sense, and the coordinate that generates it carries no data.

10.2 Why it appeared only after the hyperparameter fix

Before the fix Hes1's parameters were unidentified and pinned near zero; with $a \approx 10^{-9}$ the aH coupling was inert, the posterior was flat, and NUTS wandered it comfortably to $\widehat{R} = 1.005$ while exploring a region that meant nothing. Afterwards a is determined at 0.032, the coupling is live, and the same sampler takes 2.79 million leapfrog steps per chain and reaches $\widehat{R} = 1.76$. This is not a regression: the problem went from flat, easy and meaningless to concentrated, hard and correct.

10.3 No fixed metric can do it

The obstruction is not the conditioning at any one point but the mutual disagreement between Hessians at different points, since a mass matrix must whiten all of them at once:

system	pairwise cond$(H_i^{-1}H_j)$	mode metric	averaged metric
hiv	1.18	1.18	1.14
fn	2.1×10^3	3.8×10^2	59
lorenz	9.5×10^{10}	1.1×10^2	46
hes1	$\mathbf{1.7 \times 10^{13}}$	3.7×10^5	$\mathbf{5.2 \times 10^{11}}$

On Hes1 the averaged metric is six orders *worse* than the mode's, so the cheap alternatives are ruled out without trying them. Lorenz is the instructive near-miss: its pairwise figure is also enormous, yet a good fixed metric still reduces it to 46.

10.4 What we would try next

1. **Observe H.** The funnel is generated by a component with no data. If the experiment admits any observation of it the geometry largely dissolves. This is a change to the problem rather than to the method, and it is by some distance the cheapest item here; it is worth establishing before building machinery.

2. **Reparameterise to the profiled coordinates.** Sampling (θ, y) with $X = X^\star(\theta) + L(\theta)y$ removes the θ–X coupling by construction, and is exactly the decomposition Section 6 already computes. The obstacle is that HMC needs gradients through an argmin and a log-determinant; the implicit function theorem supplies the first, and the second requires third derivatives of U, which the structure of (1) makes available—$\nabla^2 R_a = \sqrt{b}(L_k^\top)_a \nabla^2 f$, so a third derivative is one more level on the user's vector field alone.

3. **Riemannian HMC with a SoftAbs metric**, the standard answer to a position-dependent geometry. It needs the same third derivatives and is slow per step, but it does not require the profiled construction to be correct.

4. **MCMC on the profiled parameter density.** With $p = 7$ a derivative-free sampler is entirely practical and would settle whether the importance sampling of Section 6 is adequate. It is worth being clear about what this does not do: it shares the inner Laplace approximation with the method under test, so it validates the parameter-space integration and nothing else. It is not an independent reference.

We note that the profiled posterior is structurally unaffected by this particular funnel, since it integrates the states out at each θ instead of traversing the joint geometry; on Hes1 it reports an effective sample size of 21% and $\widehat{k} = -0.06$ where NUTS fails outright. That is suggestive rather than established—without a reference we cannot score the answer—but if it holds it is an argument for the method that has nothing to do with speed.

11 Conclusion

MAGI's posterior is mostly state and a little parameter, and the two deserve different treatment. Integrating the states out exactly at each parameter value, and doing the small remaining integral directly, reaches the accuracy of a long reference chain in seconds and reports whether it has. Getting there depended on four unglamorous corrections—scaling a linear solve, integrating a benchmark accurately, restoring a prior term the derivation always contained, and fitting the Gaussian process hyperparameters on a scale-free footing—each of which was invisible in the outputs until it was looked for. The last was the largest by some distance, and the most instructive: it did not make the method fail, it made it succeed at fitting the wrong posterior, and every certificate we had built agreed that the fit was good. What caught it was plotting the trajectories.

A pattern runs through all of them. Five separate quantities in this pipeline were being compared against absolute thresholds while carrying units: the conditioning of the normal equations, the eigenvalues used to detect flat directions, the finite-difference step, the convergence test, and the agreement test between restarts. Each worked on the problem it was developed on and failed on another, and in every case the repair was the same—divide by the scale that the quantity is measured in, and the threshold becomes a statement about the posterior rather than about the units. It is a dull principle, and it accounted for more of the difficulty here than any of the approximation theory.

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